14 Extendible Hashing (Example - Photos) (ADS Notebook)

→ Extendible Hashing is a dynamic hashing method wherein directories and buckets are used to hash data.

→ The directories store address of the buckets in pointers.

→ The buckets are used to hash the actual data.

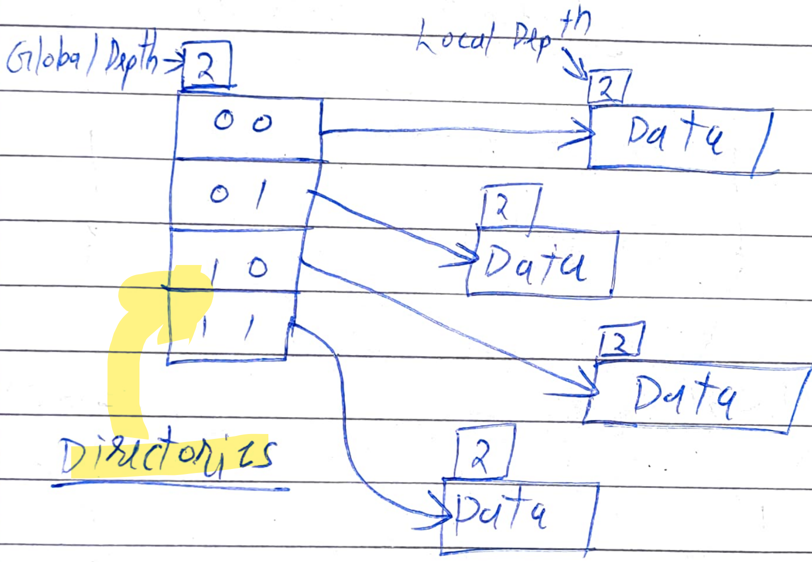


Fig. Basic structure of Extendible Hashing

→ Number of directories = $2^{\text{Global\_Depth}}$

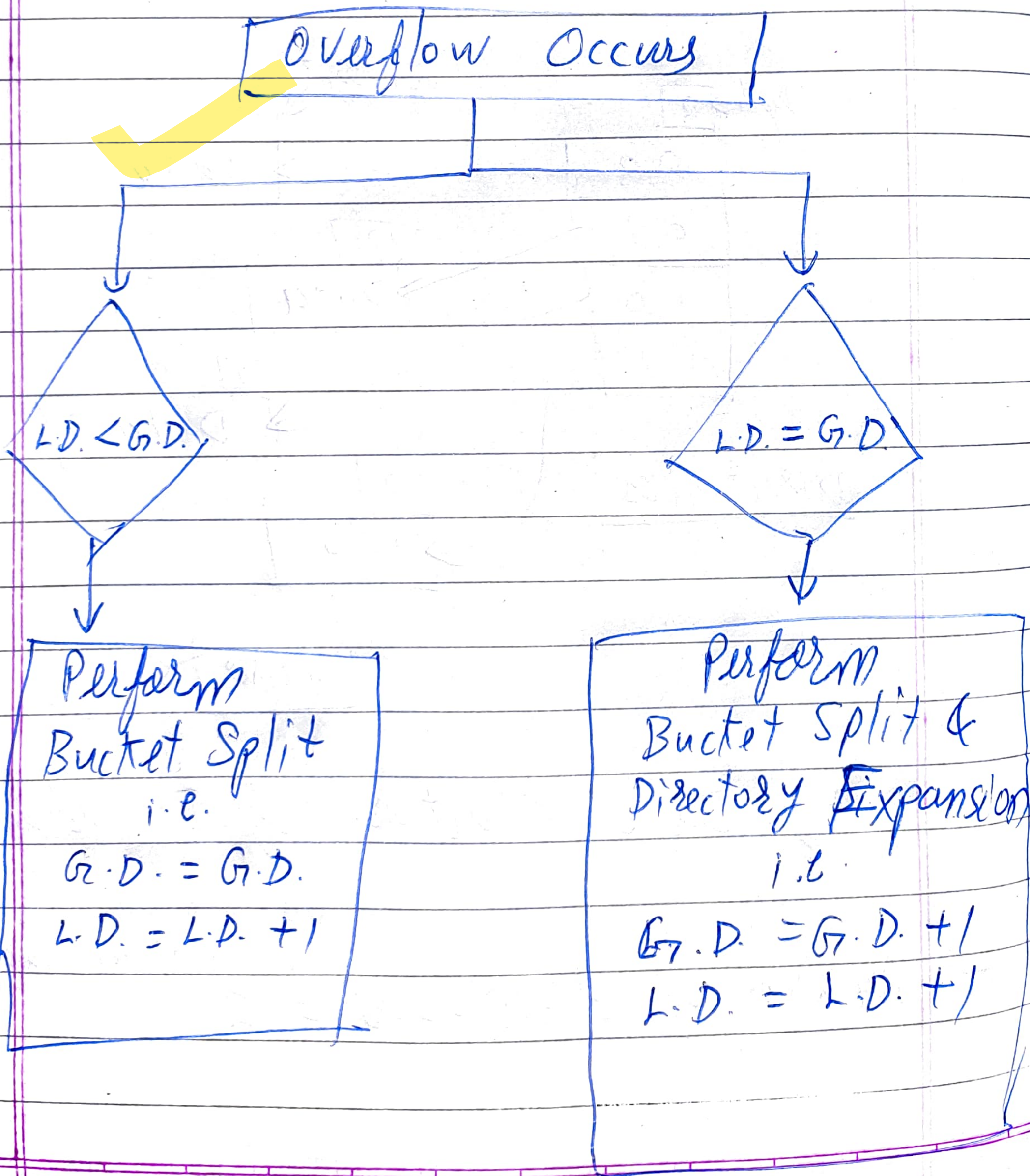
associated with directories

→ Global depth = Number of bits in directory id.

→ Local depth = Associated with buckets.

$\boxed{L.D. \leq G.D.}$

→ Bucket size : How many elements to store in one bucket.



		16	8	4	2	1
16	—	1	0	0	0	0
4	—	0	0	1	0	0
6	—	0	0	1	1	0
22	—	1	0	1	1	0
24	—	1	1	0	0	0
10	—	0	1	0	1	0
31	—	1	1	1	1	1
7	—	0	0	1	1	1
9	—	0	1	0	0	1
20	—	1	0	1	0	0
26	—	1	1	0	1	0

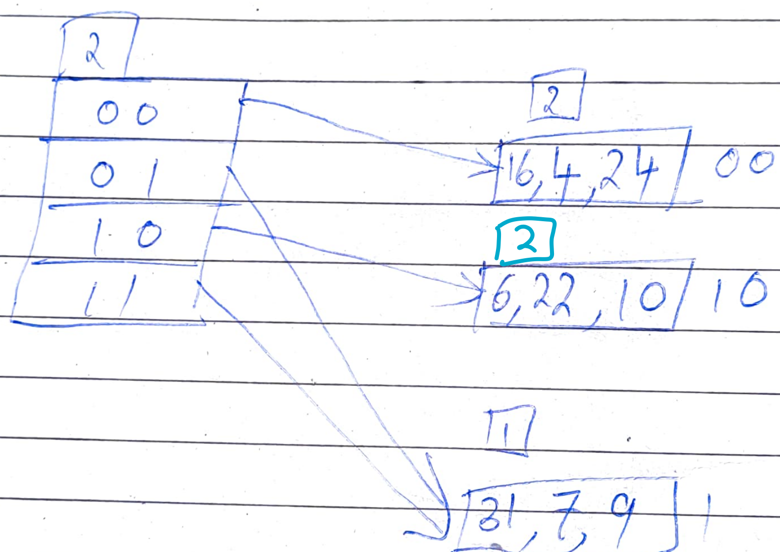
Page No.
Date

Insert

* Step 1 :- 16, 4, 6

→ Insert 32 (overflow)

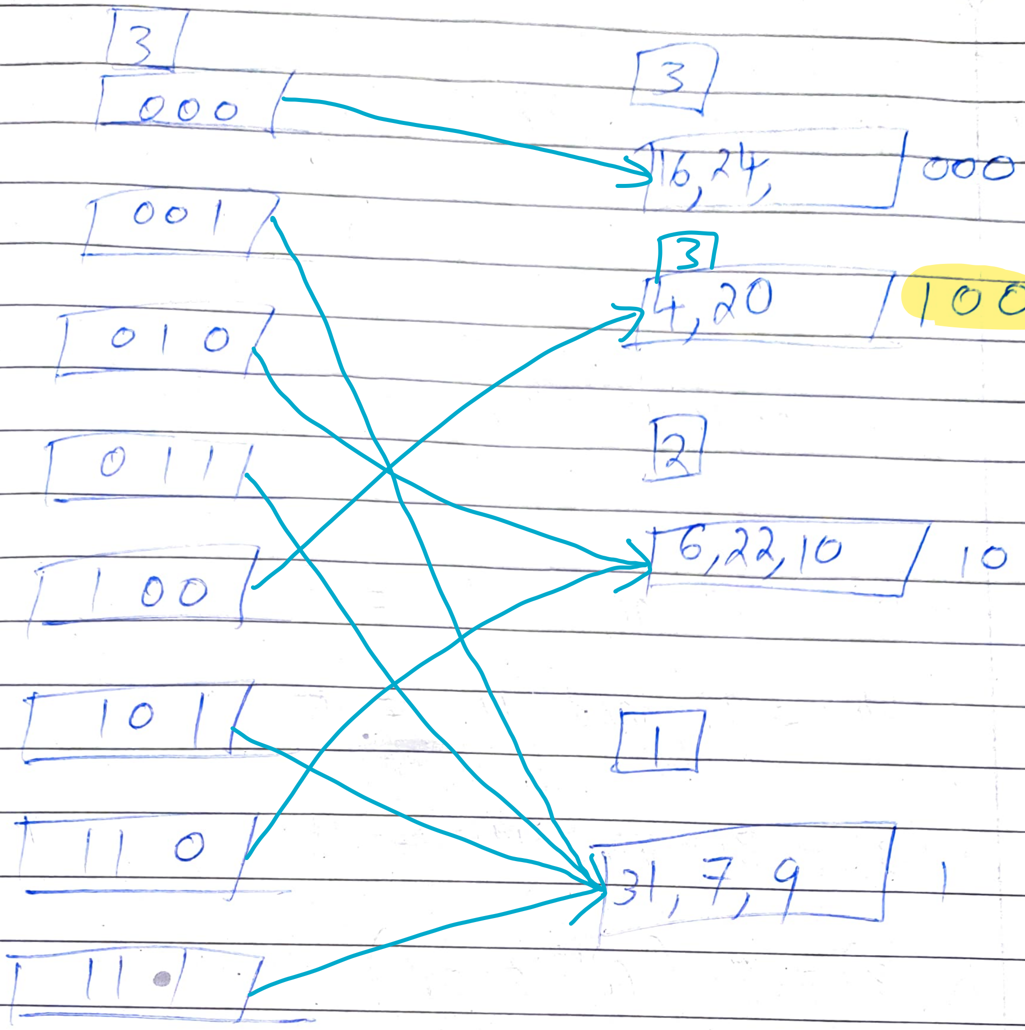
G+1 L+1



Now, insert 24, 10, 31, 7, 9

→ Insert 20 (overflow)

G+1 L+1



Insert 26 (overflow)

But, $L < G$

$\therefore L+1$